

# Vivek Dhiman

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STAFF BACKEND ENGINEER • JAVA | SPRING BOOT | ORM • DISTRIBUTED SYSTEMS • REAL-TIME DATA

## PROFESSIONAL SUMMARY

Staff backend engineer with *16+ years* designing and operating high-throughput distributed systems and real-time solutions. Delivered systems processing ~1M requests/sec at less than 10ms decision latency in cloud security (*Zscaler*) and high-volume event pipelines. Owns 10+ production microservices with 99.9%+ uptime across multi-node cloud deployments. Built and operated real time SaaS based retail backend from design to implementation. Built production AI agentic systems using Temporal, Claude API, and Context forge. Active open source contributor to Apache Druid, StarRocks, and Apache Flink — including query engine features, security hardening, and cloud storage extensions.

## CORE COMPETENCIES

|                                          |                          |                                |
|------------------------------------------|--------------------------|--------------------------------|
| Distributed Systems Design               | Data Modeling / MySQL    | Spring Boot / Security / OAuth |
| JPA and Hibernate Performance            | Real-Time Data Pipelines | AWS Cloud and CI/CD            |
| Apache Druid / Flink / Kafka / StarRocks | Docker / Kubernetes      | Team Leadership and Mentorship |

## TECHNICAL SKILLS

|                  |                                                                                                                     |
|------------------|---------------------------------------------------------------------------------------------------------------------|
| Languages        | Java 7–21 (primary), Python, C/C++, JavaScript/TypeScript, XSLT                                                     |
| Frameworks       | Spring Boot, Spring MVC, Spring Security, Spring Data JPA, Hibernate, Netty, Jetty, JUnit, TestNG, Mockito          |
| Data             | MySQL, PostgreSQL, SQLite, Apache Druid, StarRocks DB, Redis, Cassandra, ES                                         |
| Streaming        | Apache Kafka, Apache Flink, Apache Spark                                                                            |
| Cloud and DevOps | AWS (EC2, S3, CodePipeline, CloudWatch), Google Cloud, Docker, Kubernetes, Jenkins, Maven, Gradle, SonarQube        |
| Integrations     | Square POS, Stripe, Doordash, Uber Direct, Twilio SMS, Firebase, JWT, OpenAPI                                       |
| AI and Agents    | Claude API (Anthropic), Temporal (durable workflow orchestration), IBM ContextForge, human-in-the-loop agent design |

## PROFESSIONAL EXPERIENCE

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### Backend Lead Engineer — [LoyaltyLive / KidzSmart Concept](#)

Oct 2023 – Present

*Vancouver, Canada*

- ▶ Architect and own the full server-side stack for a loyalty + POS SaaS platform serving 12 active merchants, processing ~3,000 real-time orders/day across web, iOS, and Android clients.
- ▶ Built Square POS + Stripe dual-payment integration with real-time transaction sync, Doordash/Uber Direct delivery webhooks, and refund processing.
- ▶ Designed RESTful API surface for Client Portal, Admin Portal, and native mobile apps; authored full OpenAPI/Swagger specifications.
- ▶ Implemented event-driven order processing pipeline enabling reliable asynchronous workflows and horizontal scalability.
- ▶ Configured AWS CodePipeline CI/CD with CodeBuild and Secrets Manager for zero-downtime deployments.
- ▶ Designed and deployed a log-driven AI agentic pipeline using Temporal, Claude API, IBM ContextForge — detects anomalies in real-time event logs, triggers durable root cause analysis (RCA) workflows, and surfaces findings via alerts with human-in-the-loop review and approval.

### Lead Software Engineer — [EPAM Systems \(Google Fiber FDS\)](#)

Jun 2022 – Sep 2023

*Katowice, Poland*

- ▶ Embedded within Google's Fiber FDS team as primary owner of the Space Manager microservice — design, development, and delivery using Java 11 + Google Guice, Spanner DB, and GCP.
- ▶ Delivered distributed services for fiber deployment scheduling and inventory management at Google infrastructure scale, under strict reliability and code review standards.
- ▶ Collaborated with cross-functional Google engineers in an Agile environment; participated in engineering performance reviews and technical hiring.

### Senior Software Engineer — [Luxoft. DXC Company \(Citi Group\)](#)

Mar 2022 – Jun 2022

*Wrocław, Poland*

- ▶ Worked with Citi Group's Risk Management team, extending and adding new capabilities to their existing portfolio platform
- ▶ Tech Stack includes Java 11, Spring boot, Apache Storm, Kafka

### Staff Software Engineer — [Zscaler Softech India Pvt. Ltd.](#)

Jun 2019 – Feb 2022

*Mohali, India*

- ▶ Designed and operated distributed systems handling ~1M requests/sec and millions of events/sec for Zscaler's high-volume cloud security platform.

- ▶ Built ultra low-latency (<10ms) real-time decision systems: Dispatcher, LiveLogs, and Health services.
- ▶ Owned 10+ production microservices (4 built from scratch); engineered RESTful APIs over Netty/TLS/mTLS for high-throughput security traffic.
- ▶ Reduced JPA query overhead by ~80% on a complex multi-collection Product entity by eliminating Cartesian joins and redesigning the fetch strategy.
- ▶ Improved service latency and throughput using Java virtual threads, CompletableFuture async pipelines, and non-blocking I/O under high load.
- ▶ Built Kafka-based async event pipelines with Redis caching; developed production monitoring and proactive alerting systems.
- ▶ Invented and patented a Swagger tag-driven API gateway — routing rules and access policies derived automatically from OpenAPI annotations, eliminating manual gateway configuration.
- ▶ Led a 4-person engineering pod; mentored junior developers and drove Java/Spring Boot best practices across the team.

**Tech Lead, RnD — [QuarkXPress Publishing RnD India Pvt. Ltd.](#)**

Feb 2010 – May 2019

***Mohali, India***

- ▶ Owned design and development of application code across products ranging from desktop applications to web servers, building and testing features in Java 5-11, C#, XSLT, C++, Angular, JS, CSS, WPF, and HTML.
- ▶ Led a small team including one intern; conducted code reviews and gave constructive feedback to raise code quality.
- ▶ Mentored junior developers and ran internal training on Java and the Spring framework.

**Software Engineer — [Master Vision Infotech Pvt. Ltd.](#)**

Jun 2009 – Feb 2010

***Chandigarh, India***

- ▶ Built ERP solutions for multiple client organizations; first enterprise Java development exposure.

## OPEN SOURCE CONTRIBUTIONS

### Apache Druid — contributions

- ▶ Added RE2/J linear-time regex as a configurable alternative to Java regex — prevents catastrophic backtracking in query filters. [#19514]
- ▶ Added S3 Multi-Region Access Point (MRAP) and Access Point ARN support in the Druid S3 extension. [#19609]
- ▶ Added X-Error-Message trailer header to surface partial query results caused by timeout or row-limit truncation. [#16672]
- ▶ Suppressed Jackson error stack traces in HTTP responses when detailed error reporting is disabled. [#16821]
- ▶ K8s core extension, operator watch restart on stream reset [#19056]

- ▶ Additional: NPE fix in array functions [#17465], resolved Ranger extension security vulnerability; removed transitive log4j dependency [#15363], time comparison fix [#19603]

### StarRocks — contributions

- ▶ Added native HTTPS/TLS termination for StarRocks Frontend without a reverse proxy. [#60969]
- ▶ Implemented automatic LDAP group-to-database-role mapping, removing manual user provisioning on LDAP auth. [#62690]
- ▶ Fixed authentication failures for manually created users with security integration enabled. [#62693, #62696]
- ▶ Externalized Log4j configuration via system property for flexible deployment. [#69222]

### Apache Flink

- ▶ Resolved CVE-2024-47554 and CVE-2024-38374 via targeted dependency upgrades. [#26459, #26476]

### Custom Extension

- ▶ Authored druid-exact-percentile-extension — custom Druid aggregator providing exact percentile computation via reservoir sampling, with new SQL aggregation functions and configurable reservoir sizes.

## EDUCATION

**MCA — Master of Computer Applications**

2006–2009

*Chandigarh Engineering College, Landran, Punjab*